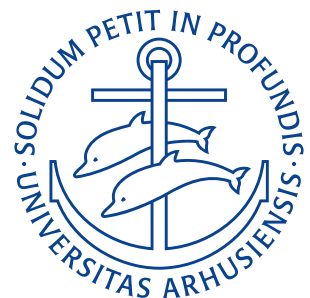


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Exercises

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Lecture 10: Internal Flows: Fully Developed Laminar Flows

24. September 2025

Problem 8.42

In the laminar flow of an oil of viscosity 1 Pa·s, the velocity at the center of a 0,3 m pipe is $4,5 \frac{\text{m}}{\text{s}}$ and the velocity distribution is parabolic. Calculate the shear stress at the pipe wall and within the fluid 75 mm from the pipe wall.

Assumptions

- Steady and incompressible flow

Derivation

As the velocity profile is parabolic the flow is fully developed. The general formula for the velocity distribution of a fully developed flow in a pipe is

$$u = -\frac{R^2}{4\mu} \left(\frac{dp}{dx} \right) \left(1 - \left(\frac{r}{R} \right)^2 \right).$$

In this case the diameter is $D = 0,3 \text{ m}$ meaning the radius is $R = \frac{D}{2} = 0,15 \text{ m}$, the flow speed is $u = 4,5 \frac{\text{m}}{\text{s}}$ and the viscosity is 1 Pa·s. At the center of the pipe $r = 0$ we have

$$u = -\frac{(0,15 \text{ m})^2}{4 \cdot 1 \text{ Pa·s}} \left(\frac{dp}{dx} \right).$$

Here we can solve for $\frac{dp}{dx}$ to get

$$\begin{aligned} \frac{dp}{dx} &= -\frac{4,5 \frac{\text{m}}{\text{s}} \cdot 4 \cdot 1 \text{ Pa·s}}{(0,15 \text{ m})^2} \\ &= -800 \frac{\text{Pa}}{\text{m}}. \end{aligned}$$

The shear stress distribution in a pipe is

$$\tau_{rx} = \frac{r}{2} \left(\frac{dp}{dx} \right).$$

At the pipe wall this is

$$\tau_{\text{wall}} = \frac{R}{2} \cdot \frac{dp}{dx} = \frac{0,15 \text{ m}}{2} \cdot 800 \frac{\text{Pa}}{\text{m}} = 60 \text{ Pa}.$$

75 mm from the pipe wall this is

$$\tau_{75 \text{ mm}} = \frac{(0,15 \text{ m} - 75 \text{ mm})}{2} \cdot 800 \frac{\text{Pa}}{\text{m}} = 30 \text{ Pa}.$$

$\begin{aligned} \tau_{\text{wall}} &= 60 \text{ Pa} \\ \tau_{75 \text{ mm}} &= 30 \text{ Pa}. \end{aligned}$

RESULT

Problem 8.13

When a horizontal laminar flow occurs between two parallel plates of infinite extent 0,3 m apart, the velocity at the midpoint between the plates is $2,7 \frac{\text{m}}{\text{s}}$. Calculate (a) the flow rate through a section 0,9 m wide, (b) the velocity gradient at the surface of the plate, (c) the wall shearing stress if the fluid has viscosity 1,44 Pas, (d) the pressure drop in each 30 m along the flow.

Assumptions

- Steady, incompressible, and fully-established flow

Derivation

In this case we must use the velocity distribution for flow between infinite plates,

$$u = \frac{a^2}{2\mu} \frac{dp}{dx} \left(\left(\frac{y}{a} \right)^2 - \frac{y}{a} \right).$$

In this particular case $a = 0,3 \text{ m}$. For the velocity at the midpoint ($y = 0,15 \text{ m}$) we get

$$V_c = \frac{(0,3 \text{ m})^2}{2 \cdot \mu} \frac{dp}{dx} \left(\left(\frac{0,15 \text{ m}}{0,3 \text{ m}} \right)^2 - \left(\frac{0,15 \text{ m}}{0,30 \text{ m}} \right) \right) = 2,7 \frac{\text{m}}{\text{s}}.$$

And therefore the pressure gradient can be found as

$$\frac{1}{2\mu} \frac{dp}{dx} = \frac{2,7 \frac{\text{m}}{\text{s}}}{(0,3 \text{ m})^2 \cdot \left(\left(\frac{0,15 \text{ m}}{0,3 \text{ m}} \right)^2 - \frac{0,15 \text{ m}}{0,30 \text{ m}} \right)} = -120 \frac{1}{\text{ms}}.$$

The average velocity is given by

$$\bar{V} = -\frac{1}{12\mu} \frac{dp}{dx} a^2.$$

Inputting known values yields

$$\bar{V} = 120 \frac{1}{\text{ms}} \cdot \frac{1}{6} \cdot (0,3 \text{ m})^2 = 1,8 \frac{\text{m}}{\text{s}}.$$

Which gives a volumetric flow rate of

$$Q = A \cdot \bar{V} = 0,3 \text{ m} \cdot 0,9 \text{ m} \cdot 1,8 \frac{\text{m}}{\text{s}} = 0,486 \frac{\text{m}^3}{\text{s}}.$$

The velocity gradient is

$$\frac{du}{dy} = \frac{a^2}{2\mu} \frac{dp}{dx} \left(\frac{2y}{a^2} - \frac{1}{a} \right) = (0,3 \text{ m})^2 \cdot \left(-120 \frac{1}{\text{ms}} \right) \cdot \left(0 - \frac{1}{0,30 \text{ m}} \right) = 36 \frac{1}{\text{s}}.$$

The shear stress at the wall is

$$\tau_{yx} = \mu \frac{du}{dy} = 1,44 \text{ Pas} \cdot 36 \frac{1}{\text{s}} = 51,8 \text{ Pa}.$$

For the pressure drop per 30 m we know that

$$\begin{aligned} \frac{1}{2\mu} \frac{dp}{dx} &= -120 \frac{1}{\text{ms}} \\ \frac{dp}{dx} &= -120 \frac{1}{\text{ms}} \cdot 2 \cdot 1,44 \text{ Pas} \\ &= -346 \frac{\text{Pa}}{\text{m}} \end{aligned}$$

Now entering the length $\Delta x = 30\text{ m}$ we get

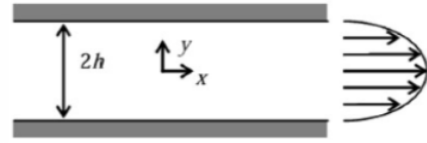
$$\Delta p_{30\text{ m}} = \frac{dp}{dx} \cdot (-\Delta x) = -346 \frac{\text{Pa}}{\text{m}} \cdot -30\text{ m} = 10,4\text{ kPa}.$$

$$\begin{aligned} Q_{0,9\text{ m}} &= 0,486 \frac{\text{m}^3}{\text{s}} \\ \left(\frac{du}{dy} \right)_{\text{wall}} &= 36 \frac{1}{\text{s}} \\ \tau_{\text{wall}} &= 51,8\text{ Pa} \\ \Delta p_{30\text{ m}} &= 10,4\text{ kPa}. \end{aligned}$$

RESULT

Problem 8.6

A fully developed and laminar flow of oil occurs between parallel plates. The pressure gradient creating the flow is $1,25 \frac{\text{kPa}}{\text{m}}$ of length and the channel half-width is $1,5 \text{ mm}$. Calculate the magnitude and length direction of the wall shear stress at the upper plate surface. Find the volume flow rate through the channel. The viscosity is $0,50 \text{ N s/m}^2$.



Derivation

The basic velocity distribution is

$$u = -\frac{a^2}{2\mu} \frac{dp}{dx} \left(1 - \left(\frac{y}{a} \right)^2 \right).$$

This corresponds to a velocity gradient of

$$\frac{du}{dy} = -\frac{a^2}{2\mu} \frac{dp}{dx} \left(\frac{2y}{a^2} \right).$$

Now the shear stress distribution can be found as

$$\tau_{yx} = \mu \frac{du}{dy} = -\frac{a^2}{2} \frac{dp}{dx} \left(-\frac{2y}{a^2} \right) = -y \frac{dp}{dx}.$$

At the outer edge where $y = h$ this becomes

$$\tau_{yx} = -h \frac{dp}{dx} = -1,5 \text{ mm} \cdot 1,25 \frac{\text{kPa}}{\text{m}} = -1,88 \text{ Pa}.$$

The volume flow rate is

$$Q = \int u dA = \int_{-h}^h u \cdot b dy = -\frac{bh^2}{2\mu} \frac{dp}{dx} \cdot \int_{-h}^h 1 - \left(\frac{y}{h} \right)^2 dy = -\frac{2h^3 \cdot b}{3\mu} \frac{dp}{dx}.$$

Now we can solve for the flow rate per unit depth $\frac{Q}{b}$ to get

$$\frac{Q}{b} = -\frac{2}{3} \frac{h^3}{\mu} \frac{dp}{dx} = -\frac{2}{3} \cdot (1,5 \text{ mm})^3 \cdot \frac{1}{0,50 \frac{\text{N s}}{\text{m}^2}} \cdot 1,25 \frac{\text{kPa}}{\text{m}} = -5,6 \frac{\text{mm}^2}{\text{s}}.$$

$$\frac{Q}{b} = -5,6 \cdot 10^{-6} \frac{\text{m}^2}{\text{s}}$$

RESULT

Lecture 11: Internal Flows: Flows in Pipes and Ducts

29. September 2025

Problem 8.20

For a turbulent flow of a fluid in 0,6 m diameter pipe, the velocity 0,15 m from the wall is $2,7 \frac{\text{m}}{\text{s}}$. Estimate the wall shear stress using the $1/7^{\text{th}}$ expression for the velocity profile.

Problem 8.24

A pipe runs for an elevation of 45 m to an elevation of 115 m. The inlet pressure is 8,5 MPa and the head loss is $6,9 \frac{\text{kJ}}{\text{kg}}$. Calculate the outlet pressure for (a) the inlet at the 45 m elevation and (b) the inlet at the 115 m elevation.

Problem 8.28

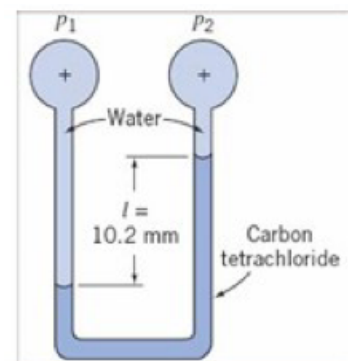
Water flows in a smooth 0,2 m diameter pipeline that is 65 m long. The Reynolds number is 10^6 . Determine the flow rate and pressure drop. After many years of use, minerals deposited on the pipe cause the same pressure drop to produce only one-half the original flow rate. Estimate the size of the relative roughness elements for the pipe.

Lecture 12: Recap

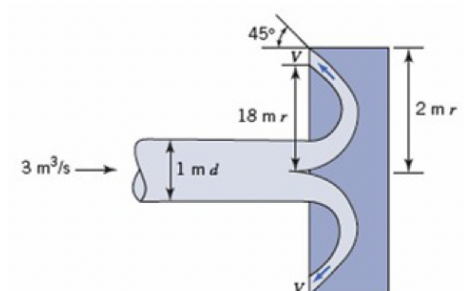
1. October 2025

Problem 3.11

Consider the two-fluid manometer shown. Calculate the applied pressure difference.

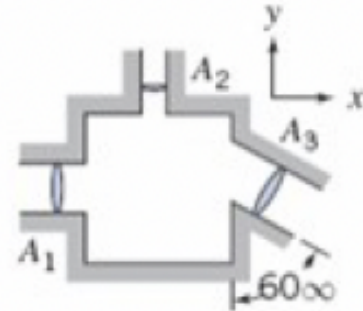
**Problem 4.10**

Find V for this mushroom cap on a pipeline.



Problem 4.4

Fluid with $1040 \frac{\text{kg}}{\text{m}^3}$ density is flowing steadily through the rectangular box shown. Given $A_1 = 0,046 \text{ m}^2$, $A_2 = 0,009 \text{ m}^2$, $A_3 = 0,056 \text{ m}^2$, $V_1 = (3 \frac{\text{m}}{\text{s}}) \hat{i}$, and $V_2 = (6 \frac{\text{m}}{\text{s}}) \hat{j}$. Determine the velocity V_3 .



Problem 5.19

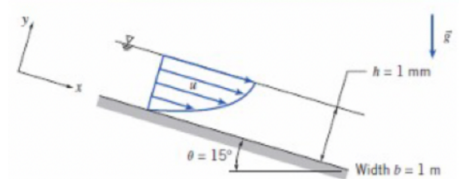
A velocity field is given by $V = 2\hat{i} - 4x\hat{j} \frac{\text{m}}{\text{s}}$. Determine an equation for the streamline. Calculate the vorticity of the flow.

Problem 5.6

The x component of velocity for a flow field is given as $u = Ax^2y^2$ where $A = 0,3 \text{ m}^{-3} \text{ s}^{-1}$ and x and y are in meters. Determine the y component of velocity for a steady incompressible flow. Determine the equation of the streamline and plot the streamline that goes through the point $(x, y) = (1, 4)$.

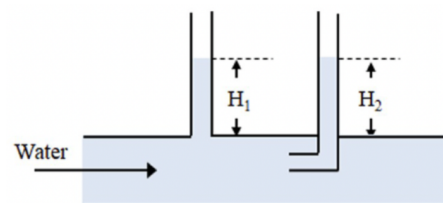
Problem 5.25

A liquid film flows on a horizontal surface due to a constant shear stress on the top surface. The liquid film is thin, the flow is fully developed, and the thickness is constant in the flow direction. Determine the velocity profile $u(x)$ and the pressure gradient $\frac{dp}{dx}$.



Problem 6.4

Water flows in a pipe at a velocity of $3 \frac{\text{m}}{\text{s}}$ and a pressure of 120 kPa absolute. Two tubes are placed in the tube as shown in the figure. Determine the height of the water in each column and the stagnation pressure in kPa.



Problem 7.22

A flat plate 1,5 m long and 0,3 m wide is towed at $3 \frac{\text{m}}{\text{s}}$ in a towing basin containing water at 20°C . The measured drag force is 14 N. Calculate the dimensions of a similar plate that will yield dynamically similar conditions in an airstream at a velocity of $18 \frac{\text{m}}{\text{s}}$, 101 kPa and 15°C . Determine the expected drag force on the plate in the air stream.

Problem 8.17

A fluid of specific gravity 0,9 flows at a Reynolds number of 1500 in a 0,3 m pipeline. The velocity 50 mm from the wall is $3 \frac{\text{m}}{\text{s}}$. Calculate the flow rate and the velocity gradient at the wall.
